

CONTROL With AUTHENTICATION For Up To Four Devices

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This circuit can be used to control up to four electrical devices through a cellphone or computer. The main advantage is that you can login from anywhere to control the device(s). If you try to login using invalid credentials 10 times, the system gets locked for the next three minutes.

Both NodeMCU v0.9 and NodeMCU v1.0 boards are based on ESP8266 Wi-Fi microchip, so the same program can be used for both. NodeMCU v0.9 consists of ESP12 chip and NodeMCU v1.0 consists of ESP12E/12F chip.

ESP8266 can be used in two different modes: station (STA) and soft access point (SAP). In STA mode, ESP8266 can connect to the Internet via another access point.

In SAP mode, ESP8266 broadcasts its own access point. Any device with Wi-Fi capability can be connected to it and used to control the device(s).

Circuit and working

The circuit diagram of the device control with authentication is shown in Fig. 1. Main controller of the circuit is NodeMCU. It gets commands from the other devices connected to it through Wi-Fi and then it changes the output pins to either on or off state according to the given command. Note that, one of the coil terminals of all the relays (RL1 through RL4) is connected to the positive rail of the 12V battery.

Here, pins D5 through D8 of NodeMCU are used as outputs. If

device1 is switched on using Wi-Fi, then pin D5 goes high, which makes pin 1 of ULN2003 high through diode D1. As a result, output pin 16 of IC2 goes low. So, RL1 gets activated and the first load gets switched on.

Similarly, if device1 is switched off using Wi-Fi, then output pin D5 goes low, pin 1 of ULN2003 goes low, and as a result pin 16 goes high. So, RL1 gets deactivated and the first load gets switched off. The same is applicable to the remaining output pins of NodeMCU, and electrical devices/loads are turned on or off, accordingly.

Software and its installation

1. Download and install the latest Arduino IDE from Arduino website.

2. Open Arduino IDE and go to File→Preferences under Settings, and paste the url http://arduino.esp8266.com/stable/package_esp8266com_index.json on Additional Board Manager URLs, as shown in Fig. 2.

3. Open Arduino IDE and go to Tools→Board→Boards Manager (Fig. 3).

4. Type ESP8266 in Search box, and install the latest version of ESP8266 by ESP8266 community.

5. After installation is complete, go to Tools→Board. Scroll down and select the correct board you are using (NodeMCU 0.9, NodeMCU 1.0, etc).

6. Go to Tools→Erase Flash and select Sketch + Wi-Fi settings.

7. Connect the board to

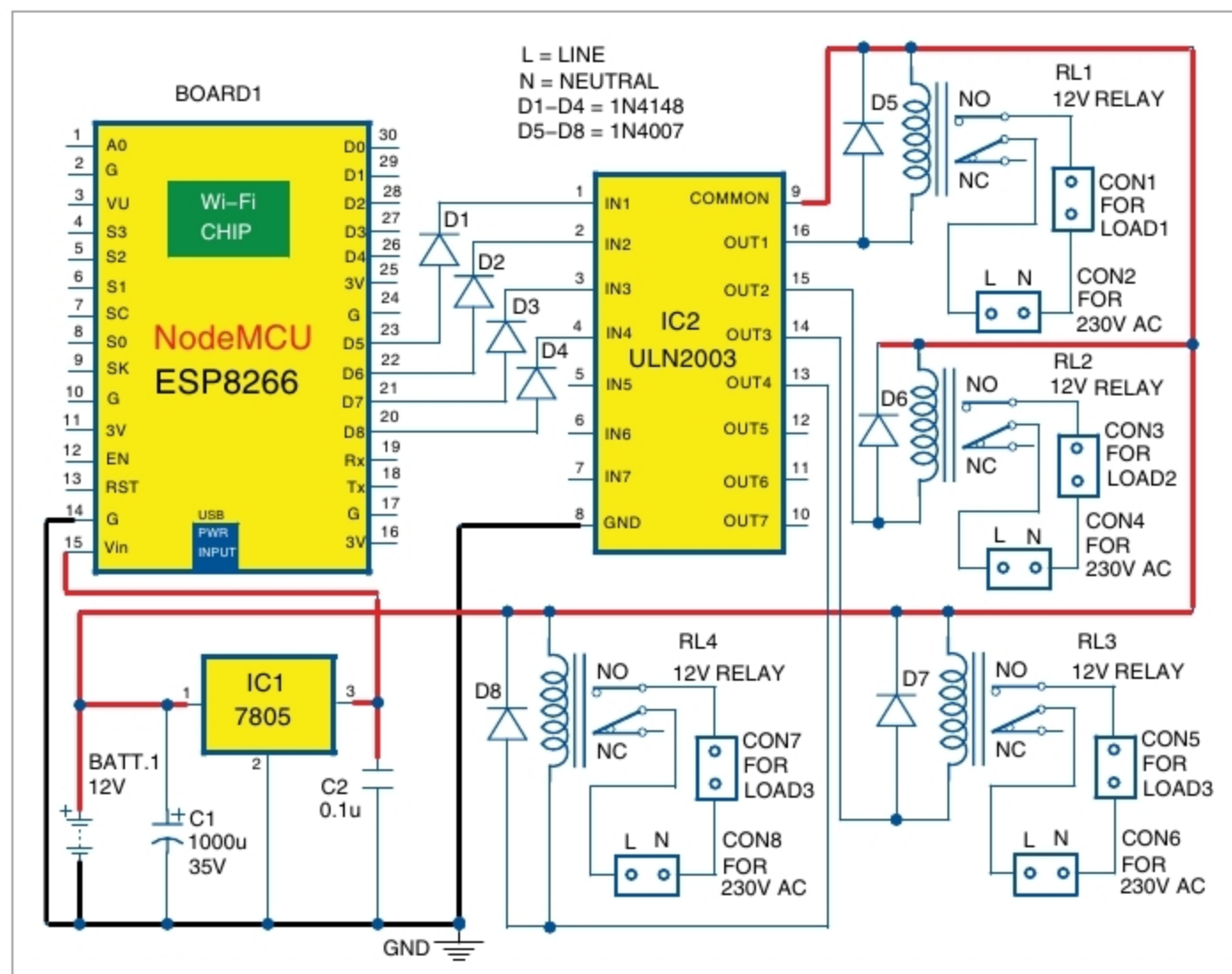


Fig. 1: Circuit diagram of device control with authentication